

Example (1-4) } 14.3
 Question (4, 8, 33) }

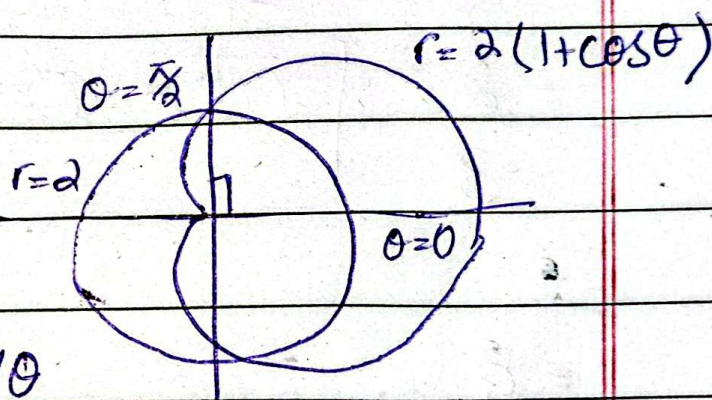
14.3

Example 1

Evaluate $\iint_R \sin \theta \, dA$ where R

is the region in the first quadrant that is outside the circle $r=2$ and inside the cardioid $r=2(1+\cos \theta)$

Sol:-



$$= \int_0^{\pi/2} \int_2^{2(1+\cos \theta)} (\sin \theta) r \, dr \, d\theta$$

$$= \int_0^{\pi/2} \left[\frac{1}{2} r^2 \sin \theta \right]_2^{2(1+\cos \theta)} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} [2(1+\cos \theta)]^2 \sin \theta - (2)^2 \sin \theta \, d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} 4(1+\cos^2 \theta) \sin \theta - 4 \sin \theta \, d\theta$$

$$= \frac{4}{2} \int_0^{\pi/2} [(1+\cos\theta)^2 \sin\theta - \sin\theta] d\theta$$

$$= 2 \int_0^{\pi/2} \left[-\frac{1}{3} (1+\cos\theta)^3 + \cos\theta \right] d\theta$$

$$= 2 \left[-\frac{1}{3} (1+\cos\frac{\pi}{2})^3 + \cos\frac{\pi}{2} \right] - \left[-\frac{1}{3} (1+\cos 0)^3 + \cos 0 \right]$$

$$= 2 \left[-\frac{1}{3} (1+0)^3 + \cancel{\cos(0)} + \frac{1}{3} \cancel{\cos(1+1)^3} - 1 \right]$$

$$= 2 \left[-\frac{1}{3} (1) + \frac{1}{3} (8) - 1 \right]$$

$$= 2 \left[-\frac{1}{3} + \frac{8}{3} - 1 \right]$$

$$= 2 \left[\frac{7}{3} - 1 \right]$$

$$= 2 \left(\frac{4}{3} \right)$$

$$= \frac{8}{3} \text{ Ans.}$$

Example 2

The sphere of radius a centered at the origin is expressed in rectangular coordinates as $x^2 + y^2 + z^2 = a^2$ and hence its eq. in cylindrical coordinates is $r^2 + z^2 = a^2$. Use this eq. and a polar double integral to find the volume of sphere.

Sol:-

$$r^2 + z^2 = a^2$$

$$z^2 = a^2 - r^2$$

$$z = \sqrt{a^2 - r^2}$$

in cylindrical coordinates
upper limit is $z = \sqrt{a^2 - r^2}$

So, the volume of the entire sphere is $V = 2 \iint_R z \, dA$

$$V = 2 \iint_R \sqrt{a^2 - r^2} \, dA$$

$$z = 0$$

$$0 = \sqrt{a^2 - r^2}$$

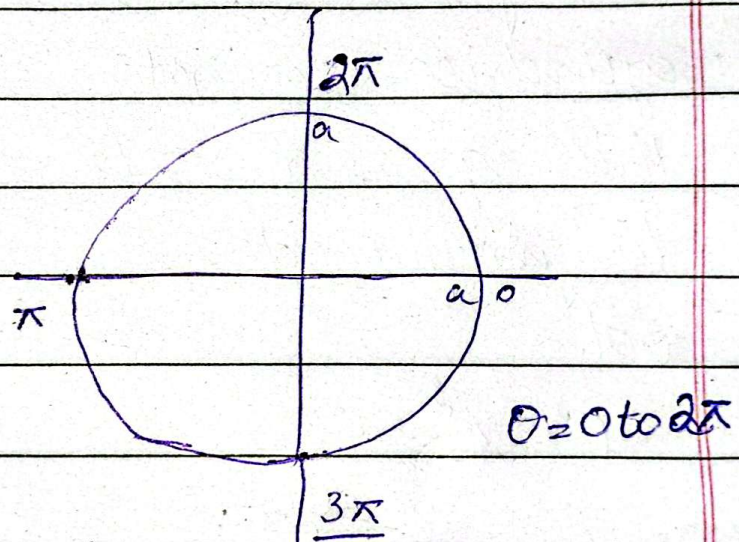
$$0 = a^2 - r^2$$

$$r^2 = a^2$$

$$r = \pm a$$

$$r = 0 \text{ to } a$$

because radius is not negative



$$= 2 \int_0^{2\pi} \int_0^a (a^2 - r^2)^{1/2} r \, dr \, d\theta$$

$$= - \int_0^{2\pi} \int_0^a (a^2 - r^2)^{1/2} (-2r) \, dr \, d\theta$$

$$= - \int_0^{2\pi} \left[\frac{(a^2 - r^2)^{3/2}}{3/2} \right]_0^a \, d\theta$$

$$= \frac{-2}{3} \int_0^{2\pi} \left[(a^2 - a^2)^{\frac{3}{2}} - (a^2 - 0^2)^{\frac{3}{2}} \right] d\theta$$

$$= \frac{-2}{3} \int_0^{2\pi} - (a^2)^{\frac{3}{2}} d\theta$$

$$= \frac{2}{3} \int_0^{2\pi} a^3 d\theta$$

$$= \frac{2}{3} a^3 \int_0^{2\pi} 1 d\theta$$

$$= \frac{2}{3} a^3 \left[\theta \right]_0^{2\pi}$$

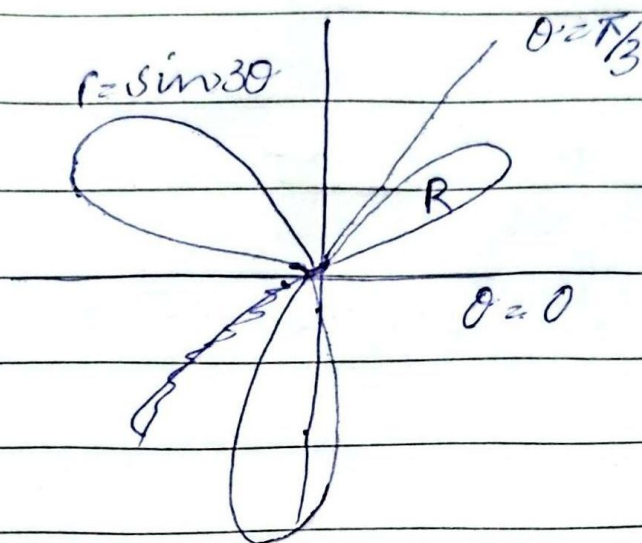
$$= \frac{2}{3} a^3 (2\pi - 0)$$

$$= \frac{4\pi a^3}{3}$$

Example 3

Use a polar double integral to find the area enclosed by the three-petaled rose $r = \sin 3\theta$

Sol:



$$A = \iint_R z \, dA = \iint_R 1 \, dA$$

multiplying 3 because of 3 petals.

$$A = 3 \int_0^{\pi/3} \int_0^{\sin 3\theta} 1 \, r \, dr \, d\theta$$

$$= 3 \int_0^{\pi/3} \left. \frac{r^2}{2} \right|_0^{\sin 3\theta} d\theta$$

$$A = \frac{3}{2} \int_0^{\pi/3} [\sin^2 3\theta - 0^2] d\theta$$

$$A = \frac{3}{2} \int_0^{\pi/3} \sin^2 3\theta d\theta$$

$$\because \cos 2\theta = 1 - 2\sin^2 \theta$$

$$2\sin^2 \theta = 1 - \cos 2\theta$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\sin^2 (3\theta) = \frac{1 - \cos 6\theta}{2}$$

$$A = \frac{3}{4} \int_0^{\pi/3} (1 - \cos 6\theta) d\theta$$

$$= \frac{3}{4} \left[\theta - \frac{\sin 6\theta}{6} \right]_0^{\pi/3}$$

$$= \frac{3}{4} \left[\left(\frac{\pi}{3} - \frac{\sin 6(\frac{\pi}{3})}{6} \right) - 0 + \frac{\sin 0}{6} \right]$$

$$= \frac{3}{4} \left(\frac{\pi}{3} - \frac{\sin 2\pi}{6} \right)$$

$$= \frac{3}{4} \left(\frac{\pi}{3} - \frac{\sin 60}{6} \right) \quad \frac{2\pi}{6} = 60^\circ$$

$$= \frac{3}{4} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$= \frac{3}{4} \left(\frac{\pi}{3} - 0 \right)$$

$$\because \frac{\sin 2\pi}{6} \approx 0$$

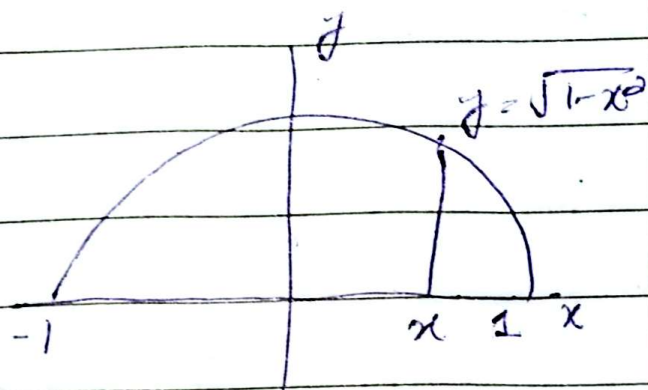
$$= \frac{3}{4} \cdot \frac{\pi}{3}$$

$$= \frac{\pi}{4} \quad \text{Ans}$$

Example 4

Use polar coordinates to
evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)^{3/2} dy dx$

Sol:-



$$= \int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)^{3/2} dy dx$$

$$= \int$$

$$y=0 \text{ to } \sqrt{1-x^2}$$

$$y=0, y=\sqrt{1-x^2}$$

$$x=1 \quad x=-1$$

for $x=-1$

$$y=0, y=\sqrt{1-x^2}$$

$$y = \sqrt{1 - (-1)^2}$$

$$y = 0$$

$$(-1, 0)$$

for $x = 1$

$$y = 0, y = \sqrt{1 - 1}$$

$$y = 0$$

$$(1, 0)$$

for $x = 0$

$$y = 0, y = \sqrt{1 - 0}$$

$$y = 1$$

$$(0, 1)$$

limit of θ

$\theta: 0 \text{ to } \pi$

limits of r

$$y = \sqrt{1 - x^2}$$

$$y^2 = 1 - x^2$$

$$x^2 + y^2 = 1$$

$$r^2 = 1$$

$$\therefore x^2 + y^2 = r^2$$

$$r = \pm 1$$

$$r = 0 \text{ to } 1$$

$$= \int_0^{\pi} \int_0^1 (r^2)^{\frac{3}{2}} r dr d\theta$$

$$= \int_0^{\pi} \int_0^1 r^3 \cdot r dr d\theta$$

$$= \int_0^{\pi} \int_0^1 r^4 dr d\theta$$

$$= \int_0^{\pi} \left. \frac{r^5}{5} \right|_0^1 d\theta$$

$$= \int_0^{\pi} \left[\frac{(1)^5}{5} - \frac{(0)^5}{5} \right] d\theta$$

$$= \int_0^{\pi} \frac{1}{5} d\theta$$

$$= \frac{1}{5} \left[(\theta) \right]_0^{\pi}$$

$$= \frac{1}{5} (\pi - 0)$$

$$= \frac{\pi}{5} \quad \text{Ans.}$$

Question 4, 8, 33 (Ex 14.3)

Question 4

Evaluate the iterated
integral:

$$\int_0^{\pi/6} \int_0^{\cos 3\theta} r \, dr \, d\theta$$

Sol:-

$$= \int_0^{\pi/6} \left[\frac{r^2}{2} \right]_0^{\cos 3\theta} d\theta$$

$$= \int_0^{\pi/6} \frac{1}{2} [(\cos^2 3\theta) - 0] d\theta$$

$$= \frac{1}{2} \int_0^{\pi/6} \cos^2 3\theta \, d\theta$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\cos^2 3\theta = \frac{1 + \cos 6\theta}{2}$$

$$= \frac{1}{4} \int_0^{\pi/6} (1 + \cos 6\theta) \, d\theta$$

$$= \frac{1}{4} \left[\theta + \frac{\sin 6\theta}{6} \right]_0^{\pi/6}$$

$$= \frac{1}{4} \left[\frac{\pi}{6} - \frac{\sin 6(\pi/6)}{6} \right] - 0$$

$$z = \frac{1}{4} \cdot \frac{\pi}{6}$$

$$\therefore \sin \frac{\pi}{6} = \frac{1}{2}$$

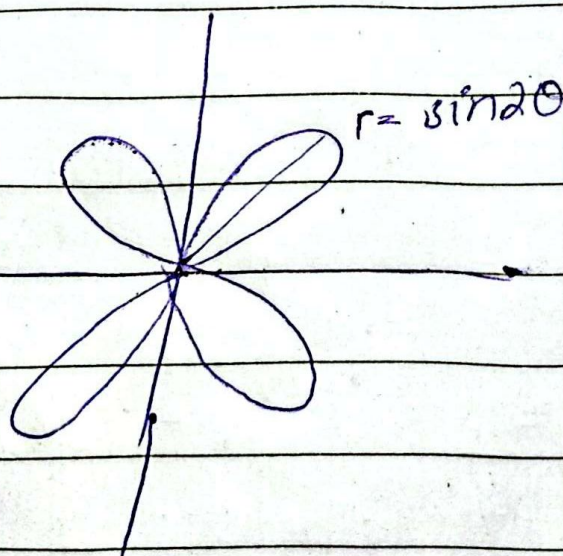
$$z = \frac{\pi}{24}$$

Question 8

Use a double integral in polar coordinates to find the area of the region described.

The region enclosed by the rose $r = \sin 2\theta$

Sol:-



$$r = 0 \text{ to } \sin 2\theta$$

limits of θ

$$r = \sin 2\theta$$

$$2\theta = \sin^{-1}(0)$$

$$\theta = 0, \frac{\pi}{2}$$

$$\theta = 0 \text{ to } \frac{\pi}{2}$$

$$A = 4 \int_0^{\pi/2} \int_0^{\sin 2\theta} r \, dr \, d\theta$$

$$= 4 \int_0^{\pi/2} \frac{r^2}{2} \Big|_0^{\sin 2\theta} d\theta$$

$$= \frac{4}{2} \int_0^{\pi/2} \sin^2 2\theta \, d\theta$$

$$= 2 \int_0^{\pi/2} \sin^2 2\theta \, d\theta$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\sin^2(2\theta) = \frac{1 - \cos 4\theta}{2}$$

$$= 2 \int_0^{\pi/2} (1 - \cos 4\theta) \, d\theta$$

$$= \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{\pi/2}$$

$$= \left[\frac{\pi}{2} - \frac{\sin 4(\pi/2)}{4} \right] - \left[0 - \frac{\sin(0)}{4} \right]$$

$$= \frac{\pi}{2} - \frac{\sin 2\pi}{4} = \frac{\pi}{2} \text{ Ans.}$$

Question 33

$$\int_0^{\sqrt{2}} \int_y^{\sqrt{4-y^2}} \frac{1}{\sqrt{1+x^2+y^2}} dx dy$$

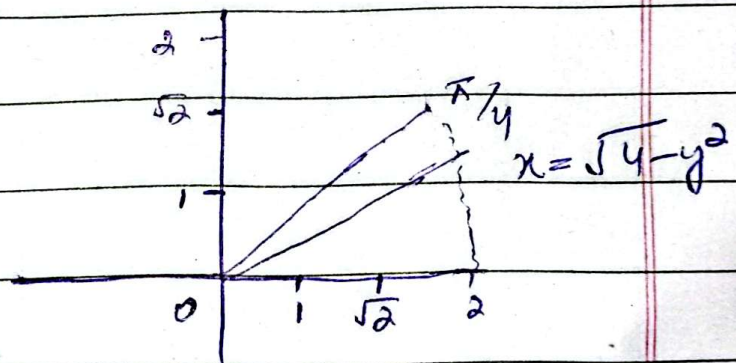
Evaluate the iterated integral by converting to polar coordinates.

Sol:-

$$x = y \text{ to } \sqrt{4-y^2}$$

$$y = 0 \text{ to } \sqrt{2}$$

$$\sqrt{2} \approx 1.4$$



$$y = 0$$

$$x = y$$

$$x = \sqrt{4-y^2} \quad \because y = 0$$

$$x = 0$$

$$x = \sqrt{4}$$

$$(0, 0)$$

$$x = 2$$

$$(2, 0)$$

$$y = \sqrt{2}$$

$$x = \sqrt{4 - (\sqrt{2})^2}$$

$$x = \sqrt{2}$$

$$(\sqrt{2}, \sqrt{2})$$

$$\theta = 0 \text{ to } \frac{\pi}{4}$$

Limits of r

$$x = \sqrt{4 - y^2}$$

$$x^2 = 4 - y^2$$

$$x^2 + y^2 = 4$$

$$r^2 = 4$$

$$r = \pm 2$$

$$r = 0 \text{ to } 2$$

$$= \int_0^{\pi/4} \int_0^2 \frac{1}{\sqrt{1+r^2}} r dr d\theta$$

$$= \int_0^{\pi/4} \int_0^2 (1+r^2)^{-1/2} r dr d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} \int_0^2 (1+r^2)^{-1/2} (2r) dr d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} \left[\frac{(1+r^2)^{-1/2+1}}{-1/2+1} \right]_0^2 d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} (1+r^2)^{1/2} \Big|_0^2 d\theta$$
$$= \frac{1}{2} \int_0^{\pi/4} (1+4)^{1/2} \Big|_0^2 d\theta$$

$$= \int_0^{\pi/4} [1+(2)^2]^{1/2} - [1+0^2]^{1/2} d\theta$$

$$= \int_0^{\pi/4} [(5)^{1/2} - 1] d\theta$$

$$= \int_0^{\pi/4} (\sqrt{5} - 1) d\theta$$

$$= (\sqrt{5} - 1) \int_0^{\pi/4} 1 d\theta$$

$$= (\sqrt{5} - 1) \theta \Big|_0^{\pi/4}$$

$$= (\sqrt{5} - 1) \left(\frac{\pi}{4} - 0 \right)$$

$$= \frac{\pi}{4} (\sqrt{5} - 1)$$